

MUSIC UNPLUGGED

The birth of compressed digital audio and the arrival of portable MP3 devices will have you humming, wherever you go

Sony's Walkman introduced us to the world of portable music. This music device could play cassettes, had a good battery life and came in a compact, convenient size. Now many decades later and various technologies past, we stand witness to another revolution with the advent of portable digital music players. And thanks to compressed audio formats like MP3, you can now carry your entire CD or tape collection and enjoy your favourite music anywhere you go. Plus, the absence of moving parts in solid-state MP3 players enables them to handle a lot of shocks and bumps, making them an ideal companion while travelling or jogging.

Whether you are looking for an MP3 player that will offer you long playtime or one which is compact enough to fit in your shirt pocket, the great news is that there is something for everyone. We test six products that aim to please our ears, with the intention of finding out the best of the bunch.

Digit Test Process

The test bed used to evaluate the portable players consisted of Windows 98 running on a Pentium III 700 MHz processor with 128 MB of SDRAM and an 8.5 GB 5,400-rpm hard drive. Test MP3 files were burned on a CD to test the CD-based MP3 players, while with solid state MP3 players, these files were uploaded into their Flash memory.

Test Methodology

We evaluated the MP3 players on features (40 per cent weightage), ease of use (10 per cent weightage), performance (20 per cent weightage), warranty and support (10 per cent weightage) and value for money (20 per cent weightage).

Features: Features like different music formats supported, presence of an FM tuner, the ability to upgrade the firmware, bundling of rechargeable batteries, etc were rated. These are the features you should be looking out for while buying a player. We also evaluated the ease with which a player lets you enjoy your music. To do this, we noted down the most important playback features, like the ability to create customised playlists, along with the facility to repeat and shuffle MP3 tracks.

Ease of use: Here we observed the ergonomics and the button placements of each MP3 player. Since these players are going to be used more outdoors than indoors, their weight, dimension and the presence of a backlit display were also noted.

Performance: We used a variety of music files (five in all) to evaluate the performance of the MP3 players and the

earphones that were bundled with them. The bass and treble test was carried to check out how purely the MP3 player produces these frequencies and whether there were any distortions. The English songs had a variety of instruments being played at the same time and they gave us an indication of the frequency response of the player. The Hindi songs, with their emphasis on vocals, gave us an indication of the clarity and the fidelity of the sounds produced.

Warranty and support: Here we took into account the warranty period for the MP3 players, the number of authorised

Compressed Audio Formats



The MP3 compression algorithm has changed the way we listen to music and the popularity of portable players that support the standard has never been greater. Fraunhofer developed the algorithm back in 1989. It was meant as a solution to cut down on the size of the very large WAV files that littered the music scene. Not only did it successfully deliver on that front, but the near CD-quality audio it offered, allowed it to capture the hearts and ears of music lovers everywhere. There are other solutions available which run neck-and-neck with the MP3 format, offering a similar solution—WMA, Ogg Vorbis, AC3, TwinVQ, RealAudio and AAC, to name a few. There are some players that support the AAC format, but its acceptance is not as wide as MP3.

Ogg Vorbis is the challenger from the open source world and delivers a comparable audio quality at zero cost. Thus, if you wish to upload your music to the Internet, you can use this compression solution to do so—royalty-free. The only problem is that it is not widely supported by MP3 players in the market. RCA Lyra was the only MP3 hardware player that supported the RealAudio format in the past but abandoned it for the more popular MP3 and WMA formats. The gaining popularity of the Windows Media Audio (WMA) format is two-fold: it offers audio quality that is comparable to an MP3 file and at a much smaller file size. It is also being aggressively marketed by Microsoft as an ideal solution to Digital Rights Management—a move that has attracted the support of a lot of music companies. As things stand, MP3 and WMA are the only formats supported by most of the portable MP3 players in the market.

